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PQ Controls INC.

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From Item No. [212N133-423] To [212N133-423]

Assembly: 212N133-423 Rev: 0

Description: 2 AXIS + GATE CAP SENSE KNOB SHORT SHAFT CAN 33H B-16809* ##

Component Item Number or Subassembly		Description / Comment	Qty Needed	Units
1	B-16697-02-SMT	HOUSING ASSY 212, DEEP DETENT W/CORNER CUTOUTS, + GATE, SS	1.0000	EA
2	B-15946-04	MODIFICATION, HOUSING, 212 DEEP DETENT W/CORNER CUTOUTS * ##	1.0000	EA
3	D-10170-04	M212 HOUSING AXIAL DETENTS DEEPENED	1.0000	EA
2	B-13802	ASSEMBLY, INNER GIMBAL WITH ENCAPSULATED MAGNET *	1.0000	EA
2	B-13803	ASSEMBLY, OUTER GIMBAL WITH ENCAPSULATED MAGNET *	1.0000	EA
2	B-12522	M212 "+" GATE PLATE* ##	1.0000	EA
3	C-10166-02	M212 SINGLE AXIS GATE PLATE	1.0000	EA
3	B-12522M	MACHINING CHARGE FOR B-12522	0.0000	EA
2	A-09354	GREASE DOW CORNING 33	0.0000	A/R
2	B-10252	STAMPING, RETAINER, BALL M212	1.0000	EA
2	A-06936-0.38S	SCREW #6 X 3/8 PLASTITE 48-2 PAN HD TAPPING SS	4.0000	EA
2	A-06936-0.75S	SCREW #6 X 3/4 PLASTITE 48-2 PAN HD TAPPING SS	2.0000	EA
2	B-10333-02	DECAL M212, X+, Y+	1.0000	EA
2	A-10728	ASSEMBLY PROCEDURE, M112/212	1.0000	
1	B-10707-133-ES	SHAFT ASSY M212, M8 THREADED KNOB, EXTRA SHORT SHAFT	1.0000	EA
2	B-16807	SHAFT 212 M8 X 1.25 THREADED, EXTRA SHORT	1.0000	EA
2	B-10088	BALL, M212	1.0000	EA
2	A-10498-03	PIN GROOVED 5/32 X 1/2 TYPE E LOW CARBON	1.0000	EA
2	A-01149-11	PIN GROOVED 5/32 X 5/16 TYPE H	1.0000	EA
2	A-10727	ASSY PROCEDURE, M212 SHAFT ASSY	1.0000	
1	B-12801	CENTERING ASSY M212, STANDARD SPRING WITH O-RING	1.0000	EA
2	C-10084	M212 CENTERING FOOT JLG#: 7020555	1.0000	EA
2	B-10506	SPRING, CENTERING M212 JLG#: 702	1.0000	EA
2	B-10179	SLEEVE, 212	1.0000	EA
2	B-10454	BOOT MOLDING M212 JLG#: 7020556	1.0000	EA
2	A-00451-09	O-RING 3/8" NITRILE DURO 2-012	1.0000	EA
		Install O-ring on Boot Collar		
1	B-15948-ES	GRIP ASSY, CAP SENSE KNOB, LARGE, NO SLEEVE	1.0000	EA
		CAP SENSE ELEC ASSY A-15893-01		
2	B-15125	UPPER HALF, CAP SENSE KNOB	1.0000	EA

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2	B-15126	LOWER HALF, CAP SENSE KNOB W/M8 INSERT	1.0000	EA
2	A-15893-01	ELEC ASSY CAP SENSE KNOB	1.0000	EA
3	A-15893	SMT ASSY CAP SENSE KNOB	1.0000	EA
4	B-15892	PCB, CAP SENSOR, CAP SENSE KNOB	1.0000	EA
4	SC-12550-05	1uF 16V CAPACITOR (KEMEK # C0805C105K4RACTU) 0805 PACKA	1.0000	EA
4	SC-14635-03	C0603C104K4RACTU 0.1uf 16V 0603 Bond	8.0000	EA
4	SC-12550-05	1uF 16V CAPACITOR (KEMEK # C0805C105K4RACTU) 0805 PACKA	1.0000	EA
4	SC-10228-01	TAJA106K010R CAPACITOR TANT 10UF 10V Bond	1.0000	EA
4	SC-12564-05	4.7UF, 16V CAP CERAMIC (0805)	1.0000	EA
4	SC-14635-02	0.01uf 50V 0603	5.0000	EA
4	SC-14907-01	18pF 16V 5% CERAMIC CAPACITOR, 0603 Bond	2.0000	EA
4	SC-12564-12	2.2nF 10% 50V ceramic cap	1.0000	EA
4	SC-12564-07	0.1UF, 50V, CAP CERAMIC (0805)	1.0000	EA
4	SZ-15157	FERRITE CHIP, 1K OHM 0.8A (0603 PACKAGE) MPZ1608S102A Bond	3.0000	EA
4	SR-12545-1501	1.5K OHM 1/10W 1% THICK FILM	2.0000	EA
4	SR-16007-5620	1/10W 0.1% 562 OHM SMT 0603 (ERA-3AEB5620V)	2.0000	EA
4	SR-12545-1200	120 OHM 1/10W 1% THICK FILM	1.0000	EA
4	SU-15156	32 BIT ARM MICROCONTROLLER, 72MHz (LQFP-48) STM32F103C8T6	1.0000	EA
4	SU-15785	IC CAP SENSE 8SOIC CY8CMBR3102	1.0000	EA
4	SU-12791-3.3BS	MIC5209-3.3YSTR 500MA LO-NOISE LDO REG	1.0000	EA
4	SU-14405	SN65HVD1050AQDRQ1 CAN TRANSCEIVER W/HIGH SPEED & EMC	1.0000	EA
4	SY-14550-08	ABM3-8.000MHZ-D2Y-T 8MHz 18pF miniature SMT Crystal	1.0000	EA
4	SZ-15276	FERRITE BEAD 600 OHM 2A (0805) TDK	1.0000	EA
3	A-07975-02-WH/RD	WH/RD WIRE 22GA TEFLON TYPE E 19/34	1.3333	FT
3	A-07975-02-BU/WH	BU/WH WIRE 22GA TEFLON TYPE E 19/34	1.3333	FT
3	A-07975-02-WH/BK	WH/BK WIRE 22GA TEFLON TYPE E 19/34	1.3333	FT
3	A-07975-02-BK/WH	BLACK/WHITE WIRE 22GA TEFLON TYPE E 19/34	1.3333	FT
3	A-07794-02	CONFORMAL COATING MS-472C BULK	0.0000	A/R
2	A-10595-0.50S	#4 X 1/2" PLASTITE SELF TAPPING 48-2 STAINLESS STEEL	2.0000	EA
2	A-00195-01S	L/W #4 EXTERNAL TOOTH SS	2.0000	EA
2	A-08113-242	LOCTITE 242 THREADLOCKER BLUE	0.0000	A/R

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<u>Component Item Number or Subassembly</u>		<u>Description / Comment</u>	<u>Qty Needed</u>	<u>Units</u>
1	A-16020-01	ELEC ASSY MCB300 HI EMC SMT SENSOR, DUAL CAN, W/ WIRES	1.0000	EA
2	A-16020-DUAL	SMT ASSY MCB300 HI EMC SMT SENSOR, STANDARD W/ DUAL CAN	1.0000	EA
3	B-16019	PCB MCB300 with High EMC	1.0000	
3	SC-09917-01	CAPACITOR .01UF 200V CERAMIC Bond	1.0000	EA
3	SC-12564-10	22uF, 20% 35V CAP CERAMIC (0805)	1.0000	EA
3	SC-12564-11	C2012X5R1V156M125AC 15uF, 20% 35V CAP	6.0000	EA
3	SC-10228-01	TAJA106K010R CAPACITOR TANT 10UF 10V Bond	1.0000	EA
3	SC-14635-01	GRM1885C1H111JA01D 110pF 16V 5% CERAMIC CAP 0603	8.0000	EA
3	SC-14635-02	0.01uf 50V 0603	23.0000	EA
3	SC-14635-03	C0603C104K4RACTU 0.1uf 16V 0603 Bond	27.0000	EA
3	SC-12564-05	4.7UF, 16V CAP CERAMIC (0805)	14.0000	EA
3	SC-14635-07	1000pf 50V 10% CERAMIC CAP 0603 Bond	8.0000	EA
3	SC-14907-01	18pF 16V 5% CERAMIC CAPACITOR, 0603 Bond	2.0000	EA
3	SC-12564-07	0.1UF, 50V, CAP CERAMIC (0805)	1.0000	EA
3	SC-14635-06	.0082uF 16V 10% CERAMIC CAP 0603 Bond	1.0000	EA
3	SC-09160-02	CL31A106KOHNNNE 10UF, 16V CAPACITOR, CERAMIC 10% Bond	2.0000	EA
3	SD-15283	SMCJ33CA 33V TVS DIODE	2.0000	EA
3	SD-15272	B560C-13-F SCHOTTKY DIODE 60V 5A Bond	1.0000	EA
3	SD-12845	B250-13-F 2A SCHOTTKY DIODE	1.0000	EA
3	SD-15273	NUP2105LT1G TVS DIODE ARRAY 27V CANBUS PROTECTION	2.0000	EA
3	SZ-15157	FERRITE CHIP, 1K OHM 0.8A (0603 PACKAGE) MPZ1608S102A Bond	14.0000	EA
3	SL-14637	CHOKE COM MODE 2x51uH 0.3A 744212510	2.0000	EA
3	SL-15278-02	CDRH6D38NP-7R4NC INDUCTOR 7.4uH 30% SMD 6038	1.0000	EA
3	SQ-14908	MOSFET N-CHANNEL 60V, BS170FTA (Diodes Inc.)	1.0000	EA
3	SR-09152-01.10	1/4W 1% 1.1 OHM SMT	1.0000	EA
3	SR-12545-1200	120 OHM 1/10W 1% THICK FILM	1.0000	EA
3	SR-12545-1002	NRC10F1002TRF 10K OHM 1/10W 1% THICK FILM	1.0000	EA
3	SR-14909-1242	1/10W 0.5% 12.4K OHM RESISTOR, RT0603DRD0712K4L (YAGEO)	13.0000	EA
3	SR-12545-2372	1/10W 1% 23.7K OHM THICK FILM	13.0000	EA
3	SR-12545-1001	1K OHM 1/10W 1% THICK FILM	14.0000	EA
3	SR-15281-6802	1/2W 1% 68K OHM ANTI SURGE SMT 0805 PACKAGE	1.0000	EA

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3	SR-15281-1202	1/2W 1% 12K OHM ANTI SURGE SMT 0805 PACKAGE	1.0000	EA
3	SR-15279-2002	1/4W 1% 20K OHM SMT 0805 PACKAGE	1.0000	EA
3	SU-15169	32 BIT ARM MICROCONTROLLER, 72MHz (LQFP-64) STM32F105R8T6	1.0000	EA
3	SU-15280	REGULATOR, SWITCHING 40V 2A	1.0000	EA
3	SU-12791-3.3BS	MIC5209-3.3YSTR 500MA LO-NOISE LDO REG	1.0000	EA
3	SU-14405	SN65HVD1050AQDRQ1 CAN TRANSCEIVER W/HIGH SPEED & EMC	2.0000	EA
3	SU-15028	SMD Hall Effect Sensor MLX90316KDC-BDG-100 MELEXIS	1.0000	EA
3	SY-14550-08	ABM3-8.000MHZ-D2Y-T 8MHz 18pF miniature SMT Crystal	1.0000	EA
2	A-07975-01-RD	RED WIRE 22GA TEFLON TYPE E 19/34	1.3333	FT
2	A-07975-01-BU	BLUE WIRE 22GA TEFLON TYPE E 19/34	1.3333	FT
2	A-07975-01-WH	WHITE WIRE 22GA TEFLON TYPE E 19/34	1.3333	FT
2	A-07975-01-BK	BLACK WIRE 22GA TEFLON TYPE E 19/34	1.3333	FT
2	A-07975-01-RD	RED WIRE 22GA TEFLON TYPE E 19/34	0.5833	FT
2	A-07975-01-BU	BLUE WIRE 22GA TEFLON TYPE E 19/34	0.5833	FT
2	A-07975-01-WH	WHITE WIRE 22GA TEFLON TYPE E 19/34	0.5833	FT
2	A-07975-01-BK	BLACK WIRE 22GA TEFLON TYPE E 19/34	0.5833	FT
2	C-10200	M212 POTTING CUP	1.0000	EA
2	A-09100	EPOXY HARDMAN 19174 PART A&B	0.0000	A/R
3	A-09100-A	EPOXY RESIN, HARDMAN 19174A	0.0000	A/R
3	A-09100-B	EPOXY HARDENER, HARDMAN 19174B	0.0000	A/R
1	A-16020-02	ELEC ASSY MCB300 HI EMC SMT SENSOR, 5V DUAL CAN, W/ WIRES	1.0000	EA
2	A-16020-5V-DUAL	SMT ASSY MCB300 HI EMC SMT SENSOR, 5V W/ DUAL CAN	1.0000	EA
3	B-16019	PCB MCB300 with High EMC	1.0000	
3	SC-10228-01	TAJA106K010R CAPACITOR TANT 10UF 10V Bond	1.0000	EA
3	SC-14635-01	GRM1885C1H111JA01D 110pF 16V 5% CERAMIC CAP 0603	8.0000	EA
3	SC-14635-02	0.01uf 50V 0603	7.0000	EA
3	SC-14635-03	C0603C104K4RACTU 0.1uf 16V 0603 Bond	11.0000	EA
3	SC-12564-05	4.7UF, 16V CAP CERAMIC (0805)	1.0000	EA
3	SC-14907-01	18pF 16V 5% CERAMIC CAPACITOR, 0603 Bond	2.0000	EA
3	SD-15273	NUP2105LT1G TVS DIODE ARRAY 27V CANBUS PROTECTION	2.0000	EA
3	SF-13340-02	SMT FUSE MFG# 1206L050 YRT	1.0000	EA

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3	SZ-15157	FERRITE CHIP, 1K OHM 0.8A (0603 PACKAGE) MPZ1608S102A Bond	1.0000	EA
3	SL-14637	CHOKE COM MODE 2x51uH 0.3A 744212510	2.0000	EA
3	SQ-14908	MOSFET N-CHANNEL 60V, BS170FTA (Diodes Inc.)	1.0000	EA
3	SR-12545-1200	120 OHM 1/10W 1% THICK FILM	2.0000	EA
3	SR-12545-1002	NRC10F1002TRF 10K OHM 1/10W 1% THICK FILM	1.0000	EA
3	SR-12545-1001	1K OHM 1/10W 1% THICK FILM	1.0000	EA
3	SU-15169	32 BIT ARM MICROCONTROLLER, 72MHz (LQFP-64) STM32F105R8T6	1.0000	EA
3	SU-12791-3.3BS	MIC5209-3.3YSTR 500MA LO-NOISE LDO REG	1.0000	EA
3	SU-14405	SN65HVD1050AQDRQ1 CAN TRANSCEIVER W/HIGH SPEED & EMC	2.0000	EA
3	SU-15028	SMD Hall Effect Sensor MLX90316KDC-BDG-100 MELEXIS	1.0000	EA
3	SY-14550-08	ABM3-8.000MHZ-D2Y-T 8MHz 18pF miniature SMT Crystal	1.0000	EA
2	A-07975-01-RD	RED WIRE 22GA TEFLON TYPE E 19/34	0.5833	FT
2	A-07975-01-BU	BLUE WIRE 22GA TEFLON TYPE E 19/34	0.5833	FT
2	A-07975-01-WH	WHITE WIRE 22GA TEFLON TYPE E 19/34	0.5833	FT
2	A-07975-01-BK	BLACK WIRE 22GA TEFLON TYPE E 19/34	0.5833	FT
2	A-07975-02-WH/RD	WH/RD WIRE 22GA TEFLON TYPE E 19/34	0.5833	FT
2	A-07975-02-BU/WH	BU/WH WIRE 22GA TEFLON TYPE E 19/34	0.5833	FT
2	A-07975-02-WH/BK	WH/BK WIRE 22GA TEFLON TYPE E 19/34	0.5833	FT
2	A-07975-02-BK/WH	BLACK/WHITE WIRE 22GA TEFLON TYPE E 19/34	0.5833	FT
2	C-10200	M212 POTTING CUP	1.0000	EA
2	A-09100	EPOXY HARDMAN 19174 PART A&B	0.0000	A/R
3	A-09100-A	EPOXY RESIN, HARDMAN 19174A	0.0000	A/R
3	A-09100-B	EPOXY HARDENER, HARDMAN 19174B	0.0000	A/R
1	B-10768-2S	ASSY KIT ELEC 212 PCB MTG DUAL AXIS *STAINLESS STEEL HARDWAR	1.0000	EA
2	A-00195-01S	L/W #4 EXTERNAL TOOTH SS	4.0000	EA
2	A-10595-0.38S	#4 X 3/8" PLASTITE SELF TAPPING 48-2 SS	4.0000	EA
1	B-12987-SS	ASSY KIT M112/M212 HANDLE WIRE MOUNTING, STAINLESS STEEL HAR	2.0000	EA
2	A-10744-08	6" CABLE TIE, W/#8 MOUNTING HOLE	2.0000	EA
2	A-06936-1.00S	SCREW #6 X 1 PLASTITE 48-2 PAN HD SS TAPPING	2.0000	EA
1	A-13862-02	SLEEVING, TECHFLEX 1/4" PET, BLACK P/N PTN0.25BK	0.5833	FT

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<u>Component Item Number or Subassembly</u>		<u>Description / Comment</u>	<u>Qty Needed</u>	<u>Units</u>
1	A-14197-06	6 POS. DEUTSCH RECEPTACLE, DTM04-6P	1.0000	EA
1	A-14198-06	6 POS DEUTSCH WEDGE FOR PINS WM-6P	1.0000	EA
1	A-14199	DEUTSCH DTM PIN, 20 AWG, 0460-202-20141	4.0000	EA
1	A-13863	SEALING PLUG DTM SERIES #0413-204-2005	4.0000	EA
1	A-14774-05	5 POS EDAC MALE HOUSING WATERPROOF (MFG #560-005-000-310)	2.0000	EA
1	A-14775	SOCKET, 22-28GA EDAC 565 SERIES (MFG #565-290-721)	8.0000	EA
1	A-14776-05	5 POS EDAC FEMALE HOUSING WATERPROOF (MFG #560-005-000-410)	2.0000	EA
1	A-14777	PIN, 22-28GA EDAC 565 SERIES (MFG #565-290-711)	8.0000	EA
1	A-14398-01	DECAL, NORTEC LABEL 1.5 x 0.51" V4975i711A5Z	1.0000	EA
1	S1169	SOFTWARE, Vermeer 212N133-422 & -423 & -395 & -397	0.0000	

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